

YANG LU

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EDUCATION

Singapore Management University

Aug 2024 - Apr 2026

Master of IT in Business (Financial Technology & Analytics)

Focus on data analytics, AI applications, and financial technology to drive business insights and digital transformation.

Singapore Management University

Aug 2022 - Apr 2024

Master of Science in Management

Focus on strategic decision-making, organisational analysis, and business problem-solving across diverse functional domains.

Singapore Institute of Management - RMIT

Jun 2019 - Dec 2021

Bachelor of Science in Construction Management

Focus on project coordination, cost management, and operational planning within large-scale construction projects.

WORKING EXPERIENCE

Talent Kraft PTE LTD – Singapore

Jan 2025 - Apr 2025

A strategy consulting firm specialising in organisational transformation, HR analytics, and talent development through data-driven and AI-enabled solutions.

Data Analytics (Intern)

- Conducted end-to-end analysis of 1,000+ employee engagement survey responses using Python (Pandas, NumPy); built multiple linear regression models and applied VIF feature reduction to identify key turnover/retention predictors with improved model reliability.
- Designed and deployed a GenAI-powered Streamlit web app (Claude API) to automate qualitative survey analysis, classifying comments into 20+ HR themes, generating sentiment labels, structured JSON insights, and interactive visualisations (Plotly, word clouds).
- Built an NLP-based job-posting evaluation pipeline to verify whether new job postings conform to the company's competency framework, using web scraping, NLP preprocessing, TF-IDF/embeddings, and cosine similarity for role-category mapping.
- Created client-facing digital deliverables using Canva, including styled reports, website mock-ups, and workshop e-brochures to support presentations and improve user experience.

CHINA CONSTRUCTION (SOUTH PACIFIC) DEVELOPMENT CO PTE LTD – Singapore

Jun 2021 - Aug 2023

Quantity Surveyor (Full-time)

- Managed monthly financial submissions and issued payment certificates via Excel to ensure accurate tracking and compliance with project milestones. Coordinated with 80+ vendors and suppliers to monitor payments, variations, and outstanding balances, and drafted contracts while negotiating pricing with subcontractors to optimise overall project costs within strict budget controls.
- Prepared and submitted client project reports and monthly claims for a S\$1.8 billion development, delivering clear financial visibility to stakeholders and enabling high-level decision support.
- Produced financial reports and forecasts, including variation orders and cost estimates, using Excel to deliver data-driven insights. Conducted quarterly cost control reviews and prepared monthly expense forecasts to support strategic budget management and identify cost-efficiency opportunities.
- Developed detailed tender documentation by quantifying the project's material quantities and deriving cost estimates from drawings, thereby providing precise financial inputs to the tendering team for negotiation and bid evaluation.

ACADEMIC PROJECTS

Text Analytics Project — Malaysia Food Tourism

Nov/2025

- Built a Python-based NLP pipeline to analyse 200K food-tourism reviews and generate insights to improve Malaysia's F&B customer experience.
- Cleaned and prepared text data by removing corrupted emoji-encoded characters, filtering duplicates, and performing tokenisation, POS tagging, stop-word removal, and lemmatisation to standardise the text.
- Conducted exploratory analysis with word clouds, frequency charts, and city-level patterns to understand customer behaviour across ten Malaysian cities.
- Developed an LDA topic model using Gensim, evaluated the coherence score to select the most meaningful number of topics, visualised topic separations with inter-topic distance maps, and manually named and assigned topics back to the entire dataset.

- Performed sentiment analysis using VADER, Multinomial Naïve Bayes, and Logistic Regression, and compared their performance to identify the most reliable method.
- Combined topic and sentiment outputs to calculate topic-level sentiment proportions and constructed a Priority Score to help identify strategic focus areas, guide service improvement plans, and support decision-making within Malaysia's F&B and tourism ecosystem.

Agricultural Yield Prediction for Bangladesh

Nov/2025

- Identified key climatic and agronomic factors influencing bumper harvests and built a data-driven framework to support agricultural planning and climate-resilient decision making.
- Cleaned and analysed more than 4,600 crop–district observations and engineered variables such as temperature and humidity stability metrics, and research-validated climate indexes.
- Segmented the dataset into cash crops, horticultural crops, and paddy crops to improve analytical relevance and enable more accurate comparisons across crop types.
- Evaluated three feature-engineering approaches that represented climate behaviour using absolute differences, region-normalized relative metrics, and research-based climatic indexes such as diurnal asymmetry and dew stability.
- Built and compared multiple models in SAS Viya including Logistic Regression, Bayesian Networks, Random Forest, Gradient Boosting, and Support Vector Machines, using F1-score, precision, recall, and AUC to assess predictive performance.
- Identified the strongest model–feature combination for each crop group and interpreted feature importance to highlight key predictors such as temperature fluctuation, humidity variability, and crop-cycle timing.

Visual Analytics for VAST Challenge 2025

Jul/2025

Project Dashboard: <https://covertreef.netlify.app>

- Conducted comprehensive Exploratory Data Analysis (EDA) and Initial Data Exploration & Analysis (IDEA) using R studio (tidyverse, ggplot2) to visualise temporal communication patterns through heatmaps, trend charts, and interactive timelines.
- Applied advanced network analysis using visNetwork and ggraph to uncover hidden relationships between vessels, officials, celebrities, and conservation groups, identifying influence pathways, communication clusters, and potential collusion.
- Built a fully interactive Shiny application enabling dynamic exploration of entities, pseudonyms, and communication patterns, supporting hypothesis testing and investigative insight generation.
- Designed and deployed a web-based interactive dashboard (R + Shiny + Netlify) with clean UI/UX to present findings in an accessible and engaging format for investigative storytelling.
- Analysed suspicious actors by combining temporal behaviours, network positions, and message-topic patterns to assess potential illegal activity and visualise evidence clearly for interpretation.

Operational Data Modeling

Apr/2025

- Designed a relational data model for a manufacturing and after-sales service environment by translating detailed business rules into a complete Entity-Relationship (ER) diagram, defining entities, attributes, cardinalities, and data integrity constraints.
- Implemented the full MySQL database using DDL by building tables, setting primary and foreign keys, defining composite keys, structuring relationships, and loading sample datasets for analysis.
- Developed analytical SQL queries to support key business insights such as monthly sales patterns, equipment condition classification, cost tracking, part-usage trends, and employee performance metrics.
- Applied advanced SQL techniques including multi-table joins, nested and correlated subqueries, CASE expressions, aggregation logic (SUM, COUNT, AVG), HAVING filters, date-time functions, string operations, and conditional logic such as IF and IFNULL.
- Created stored procedures and triggers to automate business-rule validation, enforce data consistency, and generate dynamic reports driven by user-defined input parameters.
- Strengthened technical skills in relational schema design, SQL optimisation, data quality control, and building database solutions that support end-to-end operations in a manufacturing-style service and maintenance workflow.

ADDITIONAL

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- Technical Skills: Python, R Studio (Shiny, R Markdown/HTML), Excel (VBA, Macro), SQL (MySQL), Tableau, Robotic Process Automation (blue prism), SAS Viya
 - Language Skills: English, Mandarin (Native)
 - Hobbies: Boulderling, Fishing
 - Earliest start date: April 2026